



**List of Practicals aligned with SLOs (Composite PBA)
For HSSC Annual Examination 2026 & onwards
Computer Science Curriculum (2022-23)-Scheme of Studies 2009**



Note: In the Practical-Based Assessment (PBA), questions will be taken/developed from the list of practicals provided below, aligned with the summative SLOs listed in the corresponding column.

SECTION-A (60% of practical marks — 30 Marks)

No.	List of Practicals	Aligned SLOs
1.	Write and execute simple programs that uses variables and operators with input/output handling in Python. Write and execute programs in Python that using sequence, selection, and repetition. Draw different shapes using Turtle library functions in Python. Write programs in Python using different libraries. Write and execute Python programs using function that solves a large problem by decomposing into sub problems. Write Python programs that performs some mathematical operations on a value passed to it. Write programs in Python using file handling and databases. Write and execute programs in Python using lists.	[SLO CS-11-C-01] Students should understand the importance of computer programming and applications [SLO CS-11-C-02] Students should be able to write and execute simple programs in Python. [SLO CS-11-C-03] Students should be able to draw shapes using Turtle Graphics functions in Python [SLO CS-11-C-04] Students should be able to understand the need for libraries and learn the use of some simple libraries in Python. [SLO CS-11-C-05] Students should be able to translate simple algorithms that use sequence and repetition in Python. [SLO CS-11-C-06] Students should be able to decompose a problem into sub-problems and implement those sub-problems using functions in Python [SLO CS-11-C-07] Students will determine ways of debugging their code in Python [SLO CS-12-C-02] Students should be able to use more advanced programming constructs such as data structures (lists etc.) file handling (disk IO to write to storage), and databases in Python. [SLO CS-12-C-03] Students should be able to implement complex algorithms that use lists etc. in python. [SLO CS-12-C-04] Students will determine more advanced techniques (unit tests, breakpoints, watches) for testing and debugging their code in Python.
2.	Create and test a basic prototype for business idea. Create and test a basic minimum viable product for the business.	[SLO EN-11-H-01] Students will create, test, and iterate a prototype for a business idea. [SLO EN-12-H-01] Students will create and test a minimum viable product for their business.

SECTION-B (40% of practical marks — 20 Marks)

3.	Create truth tables for expressions up to 3 inputs. Draw logic gates for a Boolean expression. Simplify the Boolean expression using Karnaugh map.	[SLO CS-11-A-01] Students will be able to understand and apply logic gates in digital systems, define and create truth tables using Boolean operators like AND, OR, NOT, NAND, XOR) and logic diagrams.
4.	Create flowchart/ pseudocode to address computational problems like: • Print the largest/smallest number • Finding the sum of all numbers • Print even/ odd numbers • Find the factorial of a number • Print the table of a number Solve computational problems like: • Sort the given list of numbers in ascending/descending order using Bubble sort / insertion sort algorithm. • Searching a given number from the list of numbers by using binary /linear search.	[SLO CS-11-B-01] Plan, develop, systematically test, and refine computational artifacts for problem-solving such as pseudocode, etc. [SLO CS-11-B-02] Apply common search, and sort algorithms
5.	Create a scatter plot/ bar chart/ pie chart/ boxplot/ line graph to show the relationship between the two variables using tools such as Excel, Google Sheets, R, or Python.	[SLO CS-11-D-01] Students will be able to relate the role and importance of model building with their real-world applications [SLO CS-11-D-02] Students will understand and explain experimental design in data science [SLO CS-11-D-03] Students will analyze pre-existing data sets to create summary statistics and data visuals (such as bar charts, pie charts, line graphs, etc.)
6.	Design a strategy for collecting data from real-life examples using: ➤ Interviews ➤ Surveys ➤ Prototypes ➤ Simulations	[SLO CS-11-G-01] Perform advanced searches to locate information and/or design a data-collection approach to gather original data (e.g. qualitative interviews, surveys, prototypes, simulations).